

Jahanavi Madireddy

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Professional Summary

Senior Data Engineer with 5+ years of experience designing scalable cloud-based data pipelines on Azure and GCP. Proficient in Python, Scala, and SQL with a deep understanding of ETL/ELT processes and orchestration tools such as Apache Airflow. Notable achievements include a 30% improvement in data processing efficiency while ensuring robust data quality, governance, and cost-effective performance.

Skills

- **Azure Services:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure Blob Storage, Azure Logic Apps, Azure Monitor
- **AWS Services:** S3, EC2, RDS, Glue, IAM, Route53, CloudWatch
- **GCP Services:** Big Query, Google Cloud Storage (GCS), Pub/Sub, Kubernetes, Docker
- **Big Data Technologies:** Apache Spark, Hive, Kafka, PySpark, MapReduce, Scala, Pig, Zookeeper, Sqoop, Flume, Oozie, Airflow
- **Hadoop Ecosystem:** HDFS, YARN, Cloudera
- **Data Warehousing:** Snowflake, Big Query, Teradata, Cosmos DB
- **Programming Languages:** SQL, Python, PL/SQL, Scala, HiveQL, Java
- **CI/CD Tools:** Jenkins, Azure DevOps
- **Reporting Tools:** Tableau, Power BI, Looker
- **Version Control:** Git, GitHub
- **Build/Automation Tools:** Maven, Ant
- **Operating Systems:** Windows, UNIX, Linux (Ubuntu, CentOS)
- **Data Engineering Practices:** ETL/ELT processes, data lakes, data modeling, data architecture, data governance, compliance, columnar storage formats, DevOps practices

Work Experience

Scripps Health

Dec 2022 - Present

Senior Data Engineer

San Diego, CA

- Designed and implemented end-to-end data pipelines using Azure Data Factory to drive efficient data ingestion, transformation, and loading (ETL) from diverse sources into a Snowflake data warehouse.
- Engineered reliable data ingestion pipelines with Azure Data Factory, Blob Storage, and Databricks to ensure seamless data transfer between systems.
- Developed and optimized ETL pipelines using Python, Scala, and SQL, boosting data processing performance by 30%.
- Automated ML pipeline processes with MLflow for experiment tracking and model deployment.
- Architected and deployed GCP-based data pipelines leveraging BigQuery, GCS, and Pub/Sub for large-scale analytics.
- Managed Kubernetes (GKE) clusters to support distributed data processing and machine learning workloads.
- Built and maintained Apache Airflow DAGs to automate data ingestion, transformation, and validation, enhancing data quality and governance.
- Coordinated migration projects using Azure Database Migration Service to transition on-premises databases to Azure with minimal downtime.
- Devised data migration strategies with Azure DMS to move SQL Server, MySQL, Oracle, and PostgreSQL systems to Azure platforms.
- Integrated diverse data sources—including databases, flat files, APIs, and cloud platforms—into a unified data warehouse using Talend.
- Orchestrated the migration of on-premises data from RDBMS and FTP servers to Azure Data Lake Storage Gen1 & Gen2.
- Unified data from on-premises databases, cloud storage, and external APIs via Azure Data Factory to build a centralized data platform.
- Utilized Python and specialized APIs to integrate external data sources, facilitating seamless data exchange.
- Developed web APIs with GraphQL for data synchronization into Aurora Postgres or DynamoDB environments.
- Crafted scalable data processing solutions using Apache Spark and Scala on Azure Databricks for high-performance transformations.
- Enhanced ETL workflows with Spark and Hive on Azure Databricks, reducing job execution times and improving processing efficiency.
- Delivered end-to-end PySpark ETL pipelines on Azure Databricks orchestrated by Azure Data Factory and scheduled via Azure automation and Tidal Scheduler.
- Scripted data transformation and cleansing tasks with T-SQL to streamline processes within SSIS packages.
- Encoded and decoded JSON objects using PySpark to manipulate data frames in Apache Spark.
- Constructed data extraction, transformation, and loading workflows with Informatica PowerCenter and Azure SQL for efficient integration.
- Developed PySpark and Spark SQL codes on Amazon EMR to execute necessary data transformations.
- Designed SSIS packages to facilitate efficient data exchanges between relational databases, flat files, and APIs.
- Managed ETL processes using Informatica Power Center and persistent data masking for data subset solutions.
- Established a data quality framework within Airflow to perform validations, consistency checks, and anomaly detection, reinforcing data reliability.

- Optimized data integration workflows using Azure Data Factory, Azure Data Lake Storage, and Databricks to improve data quality and reduce processing time.
- Designed complex data models to support data warehousing and reporting needs, ensuring optimal data structure.
- Leveraged Azure Synapse Analytics to build modern data warehousing solutions that support high-performance analytics and reporting.
- Implemented scalable data storage solutions with Azure Blob Storage to handle large volumes of structured and unstructured data.
- Utilized Polybase to load tables effectively into Azure Synapse.
- Engineered data warehousing solutions using Snowflake's multi-cluster architecture for scalable and elastic processing.
- Optimized Hadoop clusters on Azure (HDFS and YARN) to maintain high availability and scalability for big data workloads.
- Developed real-time data processing pipelines with Apache Flink to handle large-scale streaming data.
- Integrated Apache Flink with sources and sinks such as Kafka, Cassandra, HDFS, and Amazon S3 for streamlined data flow.
- Implemented data modeling and schema design using Hive on Azure Databricks to improve structured and semi-structured query performance.
- Collaborated with cross-functional teams to define data requirements and design data models in Azure Synapse Analytics to enhance data accessibility.
- Produced complex drill-down, drill-through, and parameterized reports using various reporting services to support business analytics.
- Executed data governance and security measures—including encryption, access control, and data masking—in Azure to ensure compliance with industry standards.
- Monitored and alerted on data issues using Azure Monitor and Azure Log Analytics to minimize downtime and maintain data integrity.

Charles Schwab

Aug 2021 - Nov 2022

Data Engineer

Westlake, TX

- Designed and implemented scalable data ingestion pipelines using Azure Data Factory to seamlessly ingest data from SQL databases and REST APIs.
- Developed comprehensive end-to-end data pipelines with Kafka, Spark, and Hive to enable efficient data ingestion, transformation, and analysis.
- Created Apache Airflow workflows to automate and schedule data pipelines across multiple cloud platforms, ensuring reliable operations.
- Optimized BigQuery tables with partitioning and clustering techniques, reducing query costs by 25%.
- Integrated MLflow for model versioning and performance tracking, enhancing the reproducibility of machine learning experiments.
- Developed PySpark jobs to process and transform large datasets across GCP and Azure environments efficiently.
- Streamlined data processing by automating pipelines and workflows with event-based triggers and scheduling, resulting in a 48% reduction in manual interventions.
- Leveraged Kafka and Spark Streaming to process and analyze real-time streaming data, increasing analytics capabilities by 30%.
- Supported and developed MapReduce jobs using Java API to bolster distributed data processing.
- Engineered robust data processing workflows with Azure Databricks and Spark to support distributed transformation tasks.
- Ensured data quality and integrity through rigorous validation, cleansing, and transformation processes using Azure Data Factory and Databricks.
- Developed MapReduce and Spark Python modules to support machine learning and predictive analytics within Hadoop environments.
- Utilized Spark Core and Spark SQL with Scala to expedite data processing and boost performance.
- Fine-tuned high-performance Spark jobs to manage complex data transformations, aggregations, and machine learning tasks on large datasets.
- Executed Hive scripts through Hive on Spark and Spark SQL to support ETL tasks while maintaining data integrity and pipeline stability.
- Architected and implemented a cloud-based data warehousing solution with Snowflake on Azure, leveraging its scalability and performance.
- Optimized Snowflake schemas, tables, and views to facilitate efficient data storage and retrieval for advanced analytics and reporting.
- Applied advanced partitioning, indexing, and caching techniques in Snowflake and Azure to enhance query performance and reduce processing times.
- Integrated Azure Synapse Analytics to synergize big data processing and analytics capabilities for enhanced data exploration.
- Collaborated with data analysts and stakeholders to design aligned data models within Snowflake that meet business needs.
- Implemented end-to-end data lineage and metadata management practices to ensure comprehensive governance over data transformations.
- Identified and resolved performance bottlenecks in data processing and storage layers, optimizing query execution and reducing latency.
- Utilized Python and Scala to develop custom functionalities for efficient data manipulation and integration.
- Conducted performance tuning and capacity planning exercises to ensure scalability and maximum efficiency of the data infrastructure.
- Participated actively in Agile methodologies, including daily stand-ups and planning sessions, to drive project success.

PayPal

May 2019 - Jul 2021

Junior Big Data Engineer

India

- Involved in extracting customer's big data from various data sources into Azure Blob Storage for storage in data lakes, including data from mainframes and databases.
- Integrated Databricks with various data sources, including on-premises databases, cloud-based storage solutions, and external APIs, to facilitate data ingestion and real-time updates.

- Used Azure Data Factory (ADF) to efficiently transfer data between databases and Azure Blob Storage and utilized Azure Event Hubs to stream log data from servers.
- Involved in creating streaming applications using Spark with Java on Azure HDInsight.
- Worked on data processing workflows using AWS Step Functions and Apache Airflow to orchestrate complex data pipelines and automate data processing tasks.
- Built Spark applications using Java to reduce workload on SQL and heavy coding.
- Developed various components using Java and Spring Batch for ETL batch processing jobs.
- The Hive tables created were managed with appropriate static and dynamic partitions for efficiency.
- Implemented partitioning and bucketing in Hive for better data organization.
- Developed User-Defined Functions (UDFs) in Hive.
- Implemented data quality checks and validation processes within Databricks to ensure the accuracy and reliability of data.
- Implemented the Fair Scheduler on the job tracker to allocate an appropriate number of resources to small jobs.
- Developed a data warehouse model in Snowflake for over 100 datasets using WhereScape.
- Designed and implemented Looker reports utilizing Snowflake as the data source.
- Used Azure Logic Apps workflow engine to manage interdependent jobs and automate various Hadoop tasks, including Java MapReduce, Hive, and Sqoop jobs.
- Hive is used on top of Spark for faster and better performance.
- Set up monitoring and logging for Azure Synapse to proactively identify and resolve data issues, optimizing data pipeline performance.
- Implemented metadata management and data cataloging solutions within Databricks to enhance data discoverability and lineage tracking.
- Experience with AWS services such as S3 and RDS to build scalable and cost-effective data storage solutions, optimizing performance and cost efficiency.

Education

University of North Texas

Master's, Information Systems and Technology